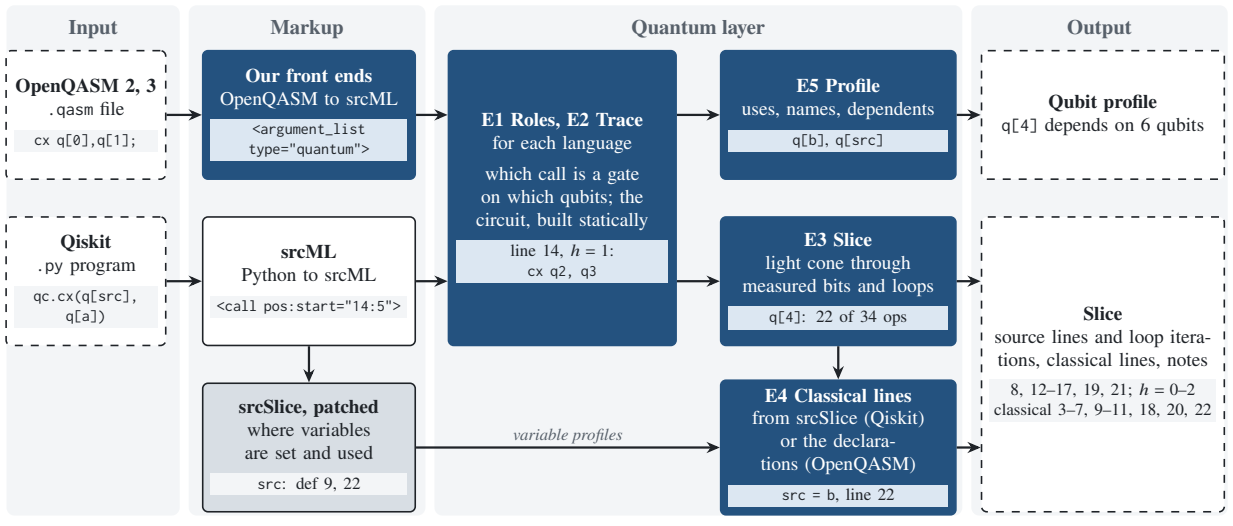

Input
Markup
Quantum layer
Output
OpenQASM 2, 3
.qasm file
cx q[0],q[1];
Qiskit
.py program
qc.cx(q[src], q[a])
Our front ends
OpenQASM to srcML
<argument_list type="quantum">
srcML
Python to srcML
<call pos:start="14:5">
srcSlice, patched
where variables are set and used
src: def 9, 22
E1 Roles, E2 Trace
for each language
which call is a gate on which qubits; the circuit, built statically
line 14, h = 1: cx q2, q3
variable profiles
E5 Profile
uses, names, dependents
q[b], q[src]
E3 Slice
light cone through measured bits and loops
q[4]: 22 of 34 ops
E4 Classical lines
from srcSlice (Qiskit) or the declarations (OpenQASM)
src = b, line 22
Qubit profile
q[4] depends on 6 qubits
Slice
source lines and loop iterations, classical lines, notes
8, 12–17, 19, 21; h = 0–2
classical 3–7, 9–11, 18, 20, 22
taken unchanged
extended and fixed
new
input or output